Dynamic programming exercises

15-E01 Rod-cutting revenue

Difficulty: Basic

p=[0,2,5,7,10,12] gives prices for lengths1..5. Compute r1..r5 and an optimal cutting of length5.

15-E02 Matrix-chain transition

Difficulty: Advanced

Matrices have dimensions2×3,3×4,4×2. Compare scalar multiplication counts for both full parenthesizations and confirm the DP optimum.

15-E03 LCS transitions and reconstruction

Difficulty: Basic

X=ABA,Y=BAA. Give the full LCS length table, optimal length, and one reconstructed LCS.

15-E04 Global alignment states and reconstruction

Difficulty: Research

X=ACGTA,Y=AGCAC. Match earns3,mismatch−2,and each insertion/deletion gap−3. Establish optimal substructure, give the full score table, reconstruct an optimum, and explain why equal-length prefixes alone are insufficient.

15-E05 Complete unbounded-knapsack DP

Difficulty: Research

Items A=(weight3,value5),B=(4,7),C=(7,12) have unlimited supply. With total weight at most13, find the exact optimum and items. Check highest value/weight greedy.

15-E06 Optimal BST with unequal failure probabilities

Difficulty: Advanced

Sorted keys k1<k2 have p1=.10,p2=.20 and failure probabilities q0=.50,q1=.10,q2=.10. Count dummy checks. Give e,w,root, reconstruct the tree, and verify expected cost by depths. Must the highest-success-probability key be root?

Dynamic programming exercises (continued)

15-E07 Failure of endpoint-only longest-simple-path states

Difficulty: Research

Directed unit-weight edges are u→z,z→v,v→z,z→w. L(a,b) stores only endpoints for longest simple paths. Find L(u,v),L(v,w), test their concatenation through v, and give a DP state preventing repeated vertices.

15-E08 Height, global weights, and two cost conventions

Difficulty: advanced

For k1<k2<k3, let p=(0.50,0.10,0.05) and q=(0.15,0.10,0.05,0.05). C counts real-key comparisons; E also counts a dummy check for each failed query.

1. Compare the balanced tree rooted at k2 with children k1,k3 and the right chain k1→k2→k3. Calculate every probability-weighted successful and unsuccessful outcome. Does smaller height guarantee smaller C?
2. Starting with C[i,i−1]=0, list w,C,root for all six nonempty intervals and reconstruct the optimum. Which values and decisions change under E?
3. If each subproblem divides its probabilities by w[i,j], can its conditional cost be added directly to the other subtree's conditional cost? Explain the missing weight factors.

15-E09 Normalized weights and all optimal solutions

Difficulty: advanced

For k1<k2<k3<k4, successful weights are P=(3,3,1,1), and gap weights are Q=(2,3,1,1,1). S is the real-key comparison weighted sum before division by total weight.

1. Use this chapter's[i,j] indices to tabulate W,S and all optimal roots for ten nonempty intervals.
2. Reconstruct every optimal tree and calculate normalized expected costs C and E. Why does dividing every weight by16 preserve the decisions?
3. Does updating one root only when c<best return every optimal tree? Is changing<to≤ enough?

> Complete your derivations before opening the separate solutions PDF.

Dynamic programming exercises (continued)

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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https://github.com/xinhi6716/computer-science-self-study

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